JCLAP · CLAP INSTRUMENT

JBreath

Breathing physical models. User Manual



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0.13.3 · Draft	CLAP · MPE · 8 voices	macOS · Windows · Linux	November 2026

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> **DRAFT**
> This is a working draft written against JBreath 0.13.3.

About this manual

Chapters 2 and 3 get you installed and playing in a few minutes. Chapter 4 tours the panel, and Chapters 5 to 7 explain how the sound is made, what each instrument is and how you blow it, which is the quickest way to understand what every control does. Chapters 8 to 14 cover every control in the order they sit on the panel, and Chapters 15 and 16 the presets, the dice and some recipes. The appendix lists every parameter with its exact range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **EMBOUCHURE**. Tags beside a parameter mean:

MOD — The host can modulate it with CLAP parameter modulation, on top of automation. Most JBreath knobs can be modulated per note.

ROUTE — A modulation route can move it, from any source in MAP or the route list (Chapter 11).

CHAPTER 01

Welcome to JBreath

An eight-voice MPE wind instrument built from physical models. You don't play a recording of a flute: you blow one.

Most wind instruments in a computer are samples, a fixed recording for each note at a few dynamics. JBreath has no samples at all. Each voice is a *bore*, a delay line that carries the sound down the tube and back with the losses and reflections of the real one, driven by an *exciter*: the reed, the lips, the air jet or the vocal folds that the player blows into. Blow harder and the reed slams shut for more of each cycle, the trumpet's wave steepens into a blare, the flute's jet jumps up a register. Nothing is crossfaded: the brightness, the growl and the split notes come out of the physics.

Six instruments share the engine: **flute**, **clarinet**, **sax**, **oboe**, **trumpet**, and an **alien** that sings words, a syllable a note. Around them sits a model of the player (the tongue, the vibrato, a hand on the bell), a second subtractive voice you can layer underneath, a modulation system read off the instrument's own breath, and a stage where every voice is drawn live as a glowing loop of its own sound, puffing smoke as it blows.

JBreath is made for expressive controllers. With an MPE keyboard or a breath controller every note has its own breath, bend and embouchure, so a chord can swell one note at a time. With a plain keyboard it still plays: the velocity blows the note.

At a glance

- **Six physical models**: an air jet, a single reed on a cylinder, a cane reed on a real alto's cone, a double reed, brass lips on a real Bb trumpet's bore, and a formant voice that sings.
- **Measured against real players.** The sax and trumpet are fitted to the University of Iowa's recordings at pp, mf and ff, harmonic by harmonic.
- **In tune everywhere.** Every note of every model is tuned to within a cent, and stays in tune as you blow softer or harder.
- **Eight voices, MPE throughout**: per-note pressure, bend and slide; breath from pressure, a breath controller (CC 2), velocity or an input you can drive from CV.
- **A player model**: five tongue styles, re-tonguing from the pressure, flutter, humanize and a vibrato from the breath, the jaw or the lip.
- **A bore you can rebuild**: length, holes, cone, bell, barrel, walls, eight materials and five mutes.
- **MAP modulation**: 22 sources, including seven read off each voice's own sound, into 32 routes and four buses.
- **The breath loops**: every voice drawn as a turning loop of its own sound, in smoke or ink.
- **87 factory presets** and a dice card that builds new patches the way a player would pick up an instrument.

BLOW, DON'T PRESS

The single biggest difference from a synth is that the breath is a continuous control, not an envelope. If your controller sends pressure, use it: hold a note and lean into it, and listen to the tone change with the level. Chapter 7 shows how to set it up.

CHAPTER 02

Installation

JBreath is a CLAP instrument. It ships in the JClap bundle alongside the other JClap plug-ins.

Requirements

Plug-in format	CLAP 1.2 (instrument, stereo out, MPE). Your host must support CLAP, for example Bitwig Studio, REAPER, FL Studio or Studio One. JBreath is CLAP only: there is no VST3 or AU version.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11.
Controller	Any MIDI keyboard. An MPE controller (Linnstrument, Seaboard, Osmose, Continuum and the like) or a breath controller on CC 2 makes it far more expressive.

Installing

01 **Download**

Grab the zip for your operating system from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. It contains every JClap plug-in.

02 **Copy the plug-in**

Copy `JBreath.clap` (a bundle on macOS) into your CLAP folder:

macOS	`~/Library/Audio/Plug-Ins/CLAP`
Windows	`C:\Program Files\Common Files\CLAP`
Linux	`~/.clap`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. JBreath appears as an instrument (a synthesizer).

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. The simplest install is from Terminal, which fetches the latest release and installs every JClap plug-in:

```
curl -fsSL https://raw.githubusercontent.com/jeffmhopkins/Jclap/main/install-macos.sh | bash
```

If you already downloaded with a browser, clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap
```

Setting up your host

Put JBreath on an instrument track and route your controller to it. For MPE, the host must pass MPE through to the plug-in: in Bitwig, JBreath declares MPE support and receives it with no extra setup; in REAPER, set the track's MIDI input to *All channels*. JBreath listens for a lower MPE zone (channel 1 the manager, 2 to 16 the notes) and follows an MPE Configuration Message if your controller sends one. It also accepts CLAP note expressions, so a host that edits per-note pressure, tuning and brightness in its piano roll plays it directly.

Ableton Live

Ableton Live is not supported. Live does not load CLAP plug-ins, and JBreath is CLAP only. If you use Live alongside a CLAP host, play JBreath in the CLAP host and bring the audio into Live.

JBreath reports its tail to the host (the release, the room and the delay), so hosts that stop processing silent tracks let it ring out.

CHAPTER 03

Quick Start

Five minutes from an empty instrument track to a horn that breathes.

01 **Load JBreath and play a note**

Put JBreath on an instrument track and hold a note. The default patch is a clarinet, *Wooden Reed*. Each note appears on the stage at the right as a glowing loop of its own sound, puffing smoke as it blows. On a plain keyboard, the velocity is how hard the note is blown.

02 **Blow it**

If your controller sends pressure (MPE or channel aftertouch) or a breath controller on CC 2, JBreath switches to it on its own within a few milliseconds: **BREATH FROM** is **AUTO**. Hold a note and lean into it. The level and the tone both follow, and past a point a reed starts to buzz. Without pressure, play the same note softly and then hard.

03 **Pick an instrument**

Click **Flute**, **Sax**, **Oboe**, **Trumpet** or **Alien** in the list on the left. Every other setting stays where it is, so this is a quick way to hear one breath through six instruments. Then try a preset: click the name in the header and choose *Clean Alto*, *Brass Plume*, *Concert Flute* or *First Contact*.

04 **Shape how it's blown**

On the exciter card, **BREATH** sets how hard the player blows at full pressure, **EMBOUCHURE** the lip and jaw, **BITE** the edge that comes in at forte, and **GROWL** a rough flutter of the throat. The graph shows the reed's valve, with a dot for each sounding note.

05 **Change the tube**

On the bore card, turn **LOSS** up for a softer, slower-speaking tube, set **TUBE** to Glass or Bamboo, or put a **MUTE** in the bell. The graph shows the bore's resonances and every note's partials on them.

06 **Map a source**

Touch **PRESS** in the header, then drag **BITE** up: a ring appears showing how far pressure moves it. Touch PRESS again to leave MAP. Now the harder you lean, the brighter the edge.

07 **Save it**

Click **SAVE** in the preset bar, type a name and press Enter. Your preset appears under *User* in the preset menu. Settings are also saved with your project automatically.

> TIP
> Every control has a tooltip, and so does every source chip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

The instrument on the left, how it's played along the bottom, and the breath loops on the right.



1. **Preset bar.** Step with the arrows, click the name for the menu, star a favorite, **SAVE** your own. The die opens the dice card.
2. **The player's hands.** Seven source chips: STRIKE, PRESS, PUSH, GLIDE, SHAKE, SLIDE, LIFT. Touch one to MAP it.
3. **PLAY / AIR.** AIR gives the breath loops the whole window.
4. **Exciter card.** EXCITER, HYBRID and (on the Alien) ALIEN tabs, with the instrument list.
5. **Air flow graph.** The valve: how the air through it answers the pressure across it.
6. **Exciter knobs.** How the instrument is blown.
7. **Bore card.** The tube: its shape, holes, material, mute, and the impedance graph.
8. **Player card.** PLAYER, VIBRATO, LFOS and BUSES tabs.
9. **Breath loops.** Every sounding voice as a turning loop, in smoke or ink.
10. **The breath.** Six source chips read off each voice's sound: LOOP, KNOT, FOLD, REGISTER, FRAY, DRAFT.
11. **LOOPS / ROUTES.** Swap the loops for the list of all 32 routes.
12. **Voice card.** VOICE, AIR, REVERB and DELAY tabs, the voice mode, and the knobs along the bottom.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated, and most knobs also accept CLAP parameter modulation from the host, per note. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
Shift + drag	Fine adjustment.
Scroll wheel	Step the value up or down.
Double-click	Reset to the default value.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number.

Each knob draws its value as an arc. When a route or the host is moving it, an outer ring shows how far, live, in the source's colour (Chapter 11).

One colour per note

Each sounding note keeps one colour across the whole panel, by the voice playing it: its loop, its smoke and its label on the stage, its partials on the bore graph, its dot on the air flow graph, the tongue and press curves, the hybrid's filter, and its place in the stereo field on the VOICE tab. Play a chord and you can follow each note through every card. The **Colours** menu above the stage picks the scheme the colours come from (Chapter 13).

Panel size

The panel is 1648 × 940 at 100%. Right-click any empty part of the panel and choose **UI Size** to scale it. The size you pick is remembered the next time the editor opens. The same menu has **Check for Updates**.

CHAPTER 05

How JBreath Works

A player, a valve and a tube. The player blows, the valve turns the steady breath into pulses, and the tube decides which pulses ring.

Signal flow

Every note gets its own voice: its own breath, its own exciter and its own bore. The voices are summed, then pass through the shared effects to the output.

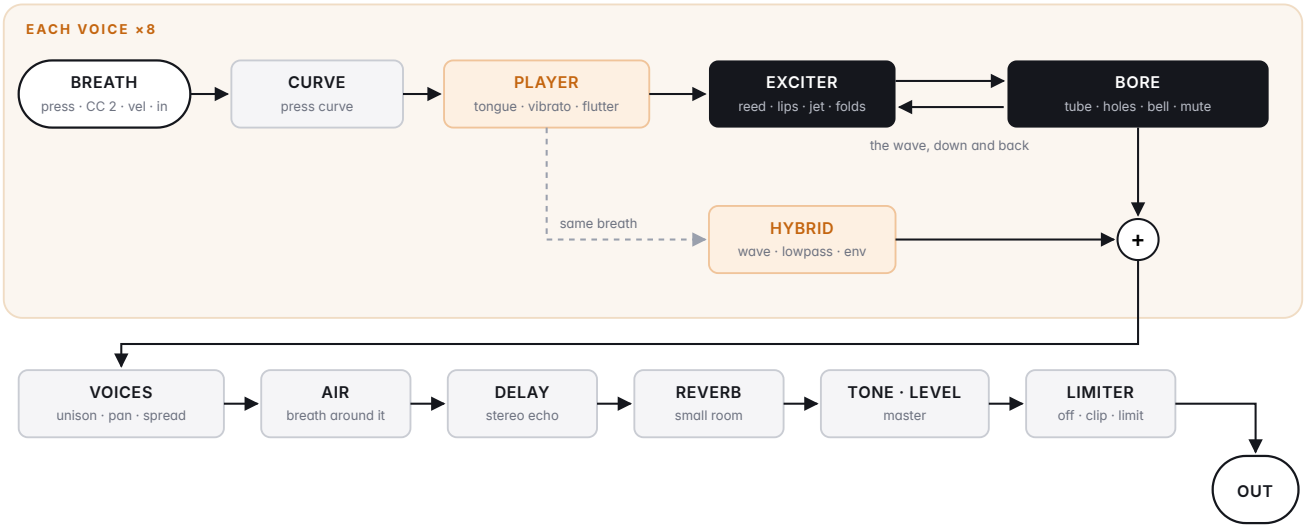


Fig. 5.1 One voice (in the shaded band, eight of them) and the shared output. The exciter and bore form a loop: the exciter pushes air into the bore, and what comes back from the bore changes how the exciter opens. The hybrid voice is blown by the same breath and mixed in after the bore.

The breath

Each note's breath starts from its source: pressure, a breath controller, the velocity or the BREATH IN parameter (Chapter 7). **CURVE** shapes it. The **tongue** then shapes the start and end of the note: a burst of pressure as the tongue lets the air go (**KICK**), the **ATTACK** and **RELEASE**, and the flutter or re-tonguing through the note. Vibrato rides on the breath, the jaw, or both. The result is a mouth pressure that changes every sample, and that is all the exciter knows about the player.

The exciter and the bore

The exciter is a valve the breath blows through: a reed, the lips, or the air jet over the flute's embouchure hole. How much air it lets through depends on the pressure across it, which depends on the sound already in the bore. The bore is a pair of delay lines carrying the wave down the tube and back, losing a little and

reflecting at the bell and the open tone holes. Each trip round the loop, the bore's resonances pull the valve into opening and closing at their frequency, and a note sounds. Nothing in this loop is a waveform or an envelope: soft playing is nearly a pure tone because the valve barely closes, and the buzz at forte is the valve slamming shut.

That is why changing the bore changes everything: **LOSS** makes notes speak slower and darker because the resonances are weaker, **HOLES** changes which highs escape, and a **MUTE** changes the bell.

In tune at any breath

The bore length for each model and note was measured offline, by playing it and correcting the length until the pitch was within a cent, and baked into the plug-in with a gain that levels each note and the softest breath that sounds it. A real player keeps a note in tune as they blow harder; JBreath measures how far each instrument drifts with the breath and corrects the bore for it. Notes past an instrument's range play an octave (or more) in.

Sample rate

The voices run at 88.2 kHz or more (twice the host's rate at 44.1 and 48 kHz) and are decimated once at the output, which keeps the reeds stable and the highs clean. At 88.2 kHz and above they run at the host's rate.

> **THE MODEL'S OWN SOUND FIRST**
> The factory preset at the head of each family (*Concert Flute*, *Wooden Reed*, *Clean Alto*, *Dusk Oboe*, *Brass Plume*, *First Contact*) is the model with a player's vibrato and a little room: the sound the instrument was tuned to. Start there before reaching for the extreme settings.

CHAPTER 06

The Instruments

Six exciters, six bores. Pick one from the list on the exciter card.

INSTRUMENT	EXCITER	BORE	WHAT GIVES IT ITS SOUND
Flute	An air jet crossing the embouchure hole, split in or out of the pipe by the sound in it.	Open at both ends: every harmonic.	The jet takes longer to reach the edge when it's slow, so blowing harder pushes it up a register. Edge noise. Nearly pure when soft, a strong 2nd and then 3rd harmonic as it's blown harder.
Clarinet	A single reed: a pressure-controlled valve, solved each sample with the pressure it makes.	A cylinder closed at the reed: odd harmonics.	The reed slamming shut at forte; the bell letting the highs out. Light on CPU.
Sax	A cane reed: a mass on a spring under the lip, pushed shut by the mouth pressure, beating on the lay at forte.	A real alto's air column, computed from its shape for each fingering: mouthpiece, conical body, nineteen tone holes, octave vents. Below the alto, the tenor, baritone and bass (tuned down to the contrabass, it plays the bass sax).	Nearly a pure tone played softly, filling in to a buzzing forte over about 12 dB, as the Iowa alto does. Low notes hold their fundamental under rich harmonics. Growl when it's pushed.
Oboe	A small, stiff double reed.	A narrow cone, built as two waveguides meeting at the reed (the bore, and a short pipe for the cone's missing tip), so its resonances are the note's harmonics; the cor anglais and lower horns below its range, down to the contrabassoon.	Nasal and reedy, with a formant near 1.8 kHz, fitted to recordings of a real oboe; darker played softly. Light on CPU.
Trumpet	The lips: a mass on a spring blown open by the mouth pressure.	A real Bb trumpet's air column for each of its seven valve combinations: mouthpiece, leadpipe, valves, Bessel bell.	The wave steepening on its way down the horn, towards a shock at forte: the brassy blare. Mellow played softly. Each tongued note scoops up into its pitch as the lips find the resonance.
Alien	Vocal folds, a pulse opening and snapping shut each period, or a bowed string.	A throat, nose and mouth: a formant synthesizer.	Every note a syllable of its words, typed or its own made-up language (Chapter 12).

Measured against real instruments

The sax and trumpet are measured against the University of Iowa's recordings of a real alto sax and Bb trumpet at pp, mf and ff, and the sax's low register against UNSW's tenor sax: how their tone darkens and how much quieter they get when played softly, how far a note starts under its pitch, and the slow 1 to 3 cent wander of a real player's pitch. The trumpet's tone through its bell is matched harmonic by harmonic without any per-note EQ: it brightens as it's blown harder because its lips close for more of each cycle. Every instrument's breath noise is a smaller share of the sound the softer it's blown.

Ranges

Under the instrument list the panel shows the range the model speaks in, for example *Speaks C1 – C8 · past that, an octave in* on the clarinet. Notes outside it are played an octave (or more) inside the range rather than failing to sound. Use **TRANSP** on the voice card, or the presets' relatives (*Piccolo*, *Bass Clarinet*, *Baritone*, *Cor Anglais*) to move the range.

Changing instrument

Choosing an instrument keeps every other setting, so the same breath, bore and player can be heard through each one. Some bore knobs only apply to some instruments (**CONE** to the sax, oboe and trumpet, **BRASS** to the trumpet): they are dimmed on the others. To start an instrument from its natural sound, load the first preset in its family instead.

CHAPTER 07

Breath and MPE

Where each note's breath comes from, and how your controller's hands reach the instrument.



Fig. 7.1 The AIR tab on the voice card, here with BREATH FROM at MIX. The graph is the press curve, with a dot for each sounding note.

Breath From

SETTING	BREATH COMES FROM	USE IT FOR
AUTO	`Velocity, until pressure or Breath In arrives`	The default. Plays from a plain keyboard and from an MPE or breath controller alike. Once pressure arrives it follows it within a few milliseconds, and velocity is then only how hard the note is tongued.
PRESS	`Pressure only`	MPE controllers and breath controllers, when you never want velocity to blow.
VEL	`Velocity only`	A plain keyboard, or a controller whose aftertouch you want to ignore.
INPUT	`The BREATH IN parameter only`	Blowing from a CV, an LFO or automation. In Bitwig, route a eurorack CV into it.
MIX	`Breath In × each note's own pressure`	The CV is the lungs for the whole chord; each finger is how much of it its note lets through. Before a note has pressure, its velocity stands in.

What counts as pressure

- **MPE pressure**: channel pressure on a note's own MPE channel.
- **Polyphonic aftertouch**: that one note's pressure.
- **Channel pressure** on a plain MIDI channel: every note on the channel.
- **A breath controller** on **CC 2**: treated as the channel's pressure.
- **CLAP pressure expressions**, from a host that edits per-note pressure in its piano roll.

The AIR tab

CONTROL	RANGE	DEFAULT	WHAT IT DOES
BREATH FROM	`Auto, Press, Vel, Input, Mix`	`Auto`	Where the breath comes from (above).
BREATH IN	`0 – 100%`	`0%`	The breath, as a parameter, for INPUT and MIX. Automate it, or modulate it from the host. MOD
CURVE	`0.30 – 3.00`	`1.00`	How pressure becomes breath. Below 1 a light touch blows harder; above 1 the breath stays soft until you press deep. MOD ROUTE
SLIDE	`0 – 100%`	`50%`	How far MPE slide (CC 74) moves the embouchure. MOD

Bend and slide

Bend bends the bore, not a pitch shifter, so a big bend sounds like a player lipping the note. MPE bend moves each note on its own, over **BEND** on the voice card (48 semitones by default, the MPE standard). The manager channel's bend moves every note, and plain MIDI bend on a non-MPE channel moves its notes by ±2 semitones. JBreath also follows the bend range your controller sets with RPN 0, and CLAP tuning expressions.

The sax and trumpet bend the way a player lips a note, on the fingering the note was played with. The trumpet keeps its valves, and the sax its whole fingering, for up to three semitones of bend either way, so a bend glides smoothly instead of stepping at each semitone, dropping an octave or losing the note. Past three semitones they change fingering as they go. Bend messages are smoothed over a few milliseconds, so a controller that sends its bend in steps doesn't make a bend buzz. A sax note lipped up two semitones is a little quieter, as a real one is.

Slide (CC 74, or a CLAP brightness expression) moves the embouchure by the SLIDE amount: on reeds, the bite; on brass, the lips' tension; on the Alien, how wide the jaw opens.

Tongue force

Whatever the tongue style, a note's velocity is how hard it's tongued: a soft touch starts gently, with little of the tongue's burst and a slower attack, and a hard one with a crisp "ta". With **FORCE** on the PLAYER tab set to **Press**, the tongue's force comes instead from how hard the pressure arrives in the note's first few milliseconds. Use it for a controller whose velocity isn't reliable, or a breath controller with no velocity at all.

Sustain and the MPE zone

The sustain pedal (CC 64) holds notes as usual, on a note's own channel or the manager channel. JBreath assumes a lower MPE zone across all 15 member channels until it receives an MPE Configuration Message; a zone of 0 makes every channel a plain MIDI channel. Reset All Controllers, All Notes Off and All Sound Off are honoured, on one channel or the whole zone.

CHAPTER 08

The Exciter Card

The instrument, and how it's blown. The HYBRID tab layers a second voice under it.



Fig. 8.1 The EXCITER tab on the clarinet. The air flow graph's curve turns over and shuts as the pressure across the reed rises; the dots are three sounding notes.

The air flow graph

The graph is the exciter's valve: how much air passes (up) for the pressure across it (along). A reed's curve rises, turns over and shuts where the reed closes; the lips' curve opens instead; the flute's jet shows where it overblows. Each sounding note is a dot riding on the curve, in its own colour, so you can see a note being blown past the top and into the buzz.

Page 1 EXCITER

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Instrument	`Flute, Clarinet, Sax, Oboe, Trumpet, Alien`	`Clarinet`	The model. Every other setting is kept when you change it.
BREATH	`25 – 150%`	`100%`	How hard the player blows at full pressure (or full velocity). Above 100% every instrument can be overblown. MOD ROUTE
EMBOUCHURE	`0 – 100%`	`50%`	Lip and jaw: the jet's angle on the flute, the bite on a reed, the lips' tension on brass. MPE slide moves it. MOD ROUTE
STIFFNESS	`0.00 – 1.00`	`0.50`	How stiff the reed or lips are (the jet's gain on a flute). Stiff is bright and quick to shut; soft is dark and lazy. MOD ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
BITE	`0 – 200%`	`100%`	The forte edge: the flute's 2nd harmonic, the reeds' edge, the brass blare. MOD ROUTE
APERTURE	`0 – 100%`	`50%`	How far the reed or lips stand open (the jet's width): wide is loud and free, narrow thin and focused. MOD ROUTE
NOISE	`0 – 200%`	`100%`	Breath noise and turbulence. Always a smaller share of the sound the softer it's blown. MOD ROUTE
KICK	`0 – 100%`	`25%`	The burst of pressure as the tongue lets the air go: the "t" at the start of a note. MOD ROUTE
GROWL	`0 – 100%`	`0%`	A rough flutter of the throat rides on the air, on any instrument; a hard-blown reed rasps more. MOD ROUTE

On the Alien, the same knobs shape its voice instead: EMBOUCHURE is the jaw, STIFFNESS the vocal folds, BITE nasal below the middle and a brassy twang above it, APERTURE the lips from rounded to a grin, and KICK how hard words start. See Chapter 12.

Page 2 HYBRID



Fig. 8.2 The HYBRID tab: a saw an octave up under *Pastoral*. The graph is its lowpass, with where the softest and hardest breath move it drawn faint.

A plain subtractive voice, blown by the same breath and layered with the instrument: a wave through a 24 dB lowpass that follows the note half way. Unison and vibrato shape it too. It has an envelope of its own, so it can also be a pad under the instrument: at full **FOLLOW** it is blown like the instrument; turn FOLLOW down and the envelope plays it from the note's strike instead, swelling in and ringing on after the breath stops.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Wave	`Sine, Triangle, Saw, Square, Pulse`	`Saw`	The oscillator.
MIX	`0 – 100%`	`0%`	How much of it. At 0% it is off. MOD ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
OCTAVE	-2 – +2 oct	0 oct	Where it plays against the instrument. MOD
DETUNE	-50 – +50 ct	+0 ct	Fine offset, for beating against the instrument. MOD ROUTE
BREATH	50 – 150%	100%	How strongly its level follows the breath. MOD
CUTOFF	80 Hz – 12 kHz	1.80 kHz	The lowpass. MOD ROUTE
RESONANCE	0 – 100%	20%	The lowpass's peak. MOD ROUTE
OPEN	0 – 100%	60%	How far the breath opens the lowpass: soft is dark, hard is bright. MOD ROUTE
ATTACK	1 ms – 10 s	5.2 ms	The envelope's attack. MOD
DECAY	1 ms – 10 s	251 ms	The envelope's decay. MOD
SUSTAIN	0 – 100%	100%	The envelope's sustain. MOD
RELEASE	1 ms – 10 s	63 ms	The envelope's release, after the note ends. MOD
FOLLOW	0 – 100%	100%	How much its level still follows the breath, against the envelope. MOD
F.ENV	-100 – +100%	0%	How far the envelope opens (or, below zero, closes) the lowpass. MOD

CHAPTER 09

The Bore

The tube the sound rings in: its shape, its holes, what it's made of, and what's in the bell.



Fig. 9.1 The bore card on *Brass Plume*. The curve is the bore's input impedance: each peak is a resonance the exciter can lock to. The lit stems are the partials of every sounding note, in each note's colour.

Reading the graph

The graph is worked out from the same loop the model runs, so every knob on the card changes it. Peaks are where the tube rings; a note's partials that land on peaks are strong, those between them weak. The resonances drawn are the loudest note's. The tabs along the top (**OPEN PIPE**, **CYLINDER**, **CONE**, **FLARE**) show the instrument's family, and the title says what that means for its harmonics.

Controls

CONTROL	RANGE	DEFAULT	WHAT IT DOES
HOLES	`Closed, Lattice 3, 5, 7, 9, Open`	`Lattice 7`	How many tone holes are open below the note. More open holes move the holes' cutoff up and radiate more highs: Closed is dark and veiled, Open bright.
TUBE	`Grenadilla, Boxwood, Maple, Bamboo, Brass, Silver, Resin, Glass`	`Grenadilla`	What it's made of. Each has its own body resonance and tilt, and soaks up more or less of every trip: wood warm and soft, metal and glass bright and ringing.
MUTE	`Open, Straight, Cup, Harmon, Plunger, Bucket`	`Open`	Something in or over the bell, on any instrument.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
TUNE	-100 – +100 ct	+0 ct	Fine tuning. MOD ROUTE
LENGTH	0.15 – 1.50 m	0.67 m	The tube below the open holes, whose own resonances ring above the holes' cutoff, most with HOLES at Closed and not at all at Open. MOD
LOSS	0 – 50%	4%	What the bore loses each trip: softer, darker, slower to speak. MOD ROUTE
HOLE CUTOFF	0.50 – 4.00 kHz	1.30 kHz	Above it the tone holes let the sound out. The pitch stays put: the filter's delay is taken out of the bore. MOD ROUTE
CONE	0 – 100%	50%	Sax and oboe: how far the cone is cut short at the reed, from towards a cylinder to a full cone. Trumpet: how conical its bore is, towards a flugelhorn. Alien: how stiff its string is. MOD ROUTE
RADIATE	0 – 100%	66%	How the bell sends the highs out. MOD
BRASS	0 – 100%	50%	Trumpet only: how far the wave steepens into a blare. MOD ROUTE
BARREL	0 – 100%	40%	The mouthpiece's cavity. MOD
WALL	Metal, Resin, Wood, Cane, Bone	Wood (50%)	How much the walls soak up. MOD
DETUNE	0 – 25 ct	+0 ct	Each note a little off, by up to this much: a section rather than one player. MOD ROUTE
MUTE AMT	0 – 100%	100%	How far the mute is in. Route slide or pressure to it to work a plunger. MOD ROUTE

Mutes

A mute does two things: it changes how the bell reflects, and it filters what comes out. **Straight**: lows cut and a nasal peak near 1.8 kHz. **Cup**: a cup over that, with a resonance near 900 Hz and the highs cut. **Harmon**: seals the bell and lets the sound out through its stem, a sharp resonance near 2 kHz over a weak fundamental: buzzy and distant. **Plunger**: a resonant lowpass that falls as it closes, the "wah", so route slide or pressure to MUTE AMT and work it like a hand. **Bucket**: felt that soaks up the highs, dark and soft. They work on every instrument: *Hat Alto*, *Felt Clarinet* and *Snake Charmer* put them on reeds.

CHAPTER 10

The Player Card

The tongue and the vibrato: how a player starts, shapes and ends each note.

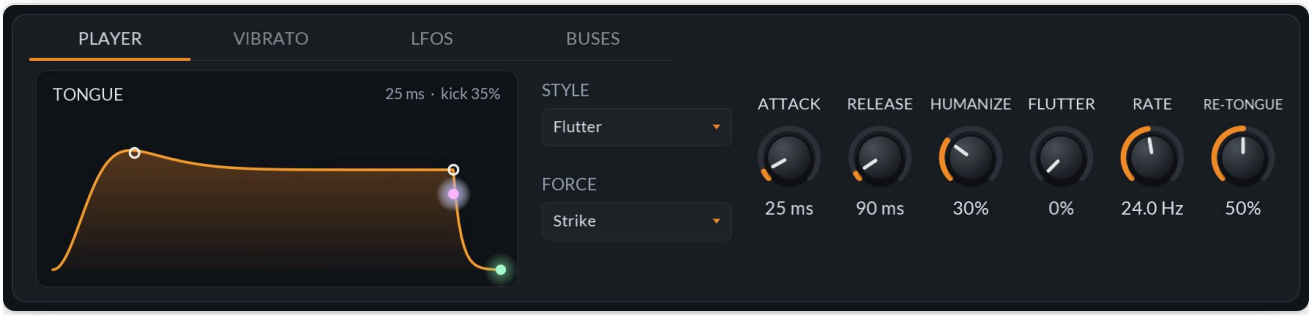


Fig. 10.1 The PLAYER tab with the Flutter style. The curve is the tongue's breath envelope; the label above it shows the attack and the kick.

Page 1 PLAYER

CONTROL	RANGE	DEFAULT	WHAT IT DOES
STYLE	`Soft, Hard, Double, Flutter, Breath`	`Hard`	How the note is tongued (below).
FORCE	`Strike, Press`	`Strike`	Where the tongue's force comes from: the velocity, or how hard the pressure arrives in the note's first few ms (Chapter 7).
ATTACK	`1 – 400 ms`	`25 ms`	How long the breath takes to arrive. MOD ROUTE
RELEASE	`5 – 2000 ms`	`90 ms`	How long it takes to stop. MOD ROUTE
HUMANIZE	`0 – 100%`	`0%`	Makes every note a little different in pitch, breath, attack and vibrato. MOD ROUTE
FLUTTER	`0 – 100%`	`0%`	Rolls the tongue through the note on any style. MOD ROUTE
RATE	`10 – 40 Hz`	`24.0 Hz`	How fast the flutter rolls. MOD ROUTE
RE-TONGUE	`0 – 100%`	`50%`	Tongues a held note again when its pressure dips quickly and comes back (below). At 0%, never. MOD ROUTE

Tongue styles

STYLE	SOUNDS LIKE	WHAT IT DOES
Soft	"da"	A gentle start, little burst.
Hard	"ta"	A crisp start with the full kick. The default.
Double	"ta-ka"	Every note tongued, even when slurred in LEGATO.
Flutter	a rolled R	Rolls the tongue through the note, at least half way whatever FLUTTER is set to.

STYLE	SOUNDS LIKE	WHAT IT DOES
Breath	"hhh"	No tongue at all: the note starts from the air alone.

Re-tonguing

A wind player repeats a note without changing fingers by tonguing it again. With pressure, JBreath does the same: when a held note's pressure falls under a dip within 150 ms and comes at least half way back within 300 ms more, the note is tongued again, as hard as the pressure came back. A slow diminuendo and swell never counts. Turned up, a lighter dip is enough; at 0% it never happens. On a single key you can now play repeated notes the way a player does on one fingering.

Page 2 VIBRATO

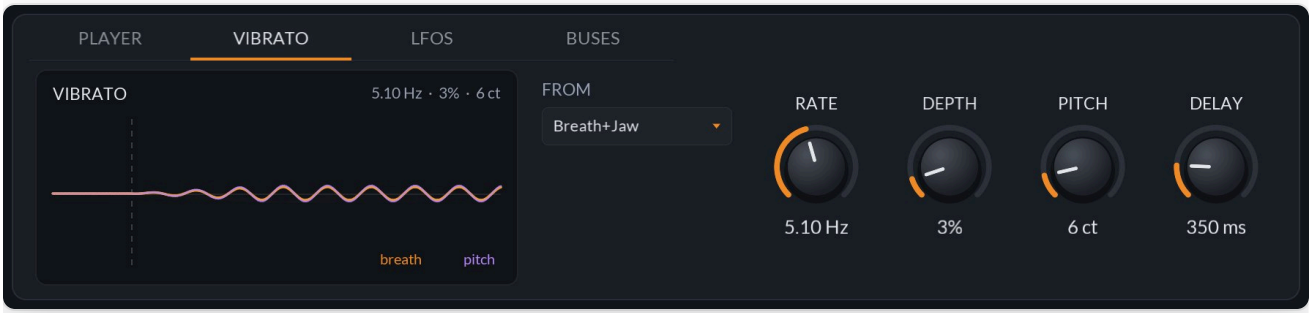


Fig. 10.2 The VIBRATO tab, from the breath and the jaw together. The graph shows how it grows in after the delay.

Vibrato grows in after **DELAY**, and its rate and depth wander a little as a player's do.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
FROM	`Breath, Jaw, Breath+Jaw, Lip`	`Breath`	Where it comes from: the diaphragm (level and colour), the jaw (the embouchure, so pitch and colour), both, or the lip (pitch alone).
RATE	`2 – 9 Hz`	`5.10 Hz`	How fast. MOD ROUTE
DEPTH	`0 – 30%`	`5%`	How deep, from FROM. MOD ROUTE
PITCH	`0 – 50 ct`	`0 ct`	Pitch vibrato in cents, on top. MOD ROUTE
DELAY	`0 – 2000 ms`	`350 ms`	How long into the note before it grows in. MOD ROUTE

The LFOS and BUSES tabs are part of the modulation system, in Chapter 11.

CHAPTER 11

Modulation

MAP any source onto any knob with a touch and a drag. Thirteen of the sources are your hands and the instrument's own breath.

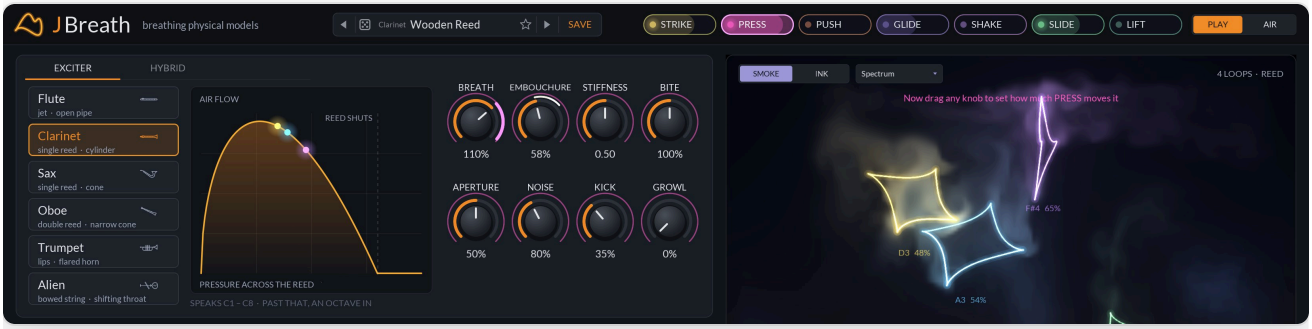


Fig. 11.1 MAP with PRESS armed. Every knob that can be routed shows a ring in PRESS's colour; dragging one sets how far pressure moves it.

MAP

01 **Arm a source**

Touch a source chip: one of the hands in the header, or one of the breath sources under the loops. The panel shows a hint and every routable knob gets a ring.

02 **Drag a knob**

Dragging now sets how far the source moves that knob, up or down, instead of the knob's value. The ring shows the reach. Double-click the knob to flip the route's sign; right-click it to remove the route.

03 **Leave MAP**

Touch the chip again. The knob's outer ring keeps moving with the source, per voice, as you play.

Outside MAP, right-click a knob to unmap every route into it. Hover a chip to read what its source is. Everything works per voice: PRESS on one note moves only that note.

The sources

New in 0.12.0: **PUSH** and **SHAKE** in the header, and **FRAY** in place of Vapor. Every breath source now runs 0 to 1 on every instrument, so a route sounds the same whichever instrument you pick.

SOURCE	RANGE	WHAT IT FOLLOWS
THE PLAYER'S HANDS · IN THE HEADER		

SOURCE	RANGE	WHAT IT FOLLOWS
STRIKE	0 – 1	The note's velocity.
PRESS	0 – 1	Its pressure.
PUSH	0 – 1	How fast the pressure is rising: an accent, falling away once the finger stops pushing. Full at the whole range in 0.1 s.
GLIDE	-1 – +1	Its pitch bend, full at two semitones whatever the bend range.
SHAKE	0 – 1	How much the finger is moving the pitch: its vibrato, or a bend on its way. Full at a quarter of a semitone.
SLIDE	0 – 1	Its slide (CC 74).
LIFT	0 – 1	Its release velocity, moving only once the note is let go.
THE BREATH · UNDER THE LOOPS, READ OFF EACH VOICE		
LOOP	0 – 1	How hard it blows, from the lightest breath that sounds to the instrument's full breath, with its attack and release.
KNOT	0 – 1	How bright it is for its note: the share of the sound above its 4th harmonic. Holds as the note dies away.
FOLD	0 – 1	How far into the top of its breath, from mezzo forte to full: towards where it overblows.
REGISTER	0 – 1	How high it plays in the instrument's range.
FRAY	0 – 1	How far the sound is from a steady tone: breath noise, growl, flutter, the attack or a split note, whatever frays the loop. Holds as the note dies away.
DRAFT	-1 – +1	A slow wandering draught of its own, carried from note to note.
GENERATORS AND BUSES · IN THE ROUTE LIST AND THE LFOS AND BUSES TABS		
LFO 1, LFO 2	-1 – +1	Two LFOs (below).
SWELL	0 – 1	One contour over its time from each note's start (below).
KEY	-1 – +1	The note, around middle C.
RANDOM	-1 – +1	A fresh value for each note.
BUS 1 – 4	-1 – +1	The buses' outputs (below).

> **BREATH SOURCES CAN FEED BACK**
> LOOP, FOLD, KNOT and FRAY are measured from the breath and the sound, so routed onto **BREATH** (or anything that changes how hard it blows) they feed back. That can be useful: FOLD onto BREATH latches a note once it's overblown, and LOOP onto BREATH at −100% flattens the dynamics. If a patch runs away, look for a breath source routed to Breath.

Destinations

A route can move any knob tagged ROUTE in this manual: the exciter, the player and vibrato, the bore's Tune, Loss, Hole Cutoff, Cone, Brass, Detune and Mute Amount, the hybrid voice, the LFO rates and Swell time, Transpose, Glide, Curve, Pan, Unison Detune, Air and Level, every Alien knob, Re-Tongue, and the

four buses' Amount. Several routes into one knob add up, and the result is kept within the knob's range. Host modulation is added first.

The route list

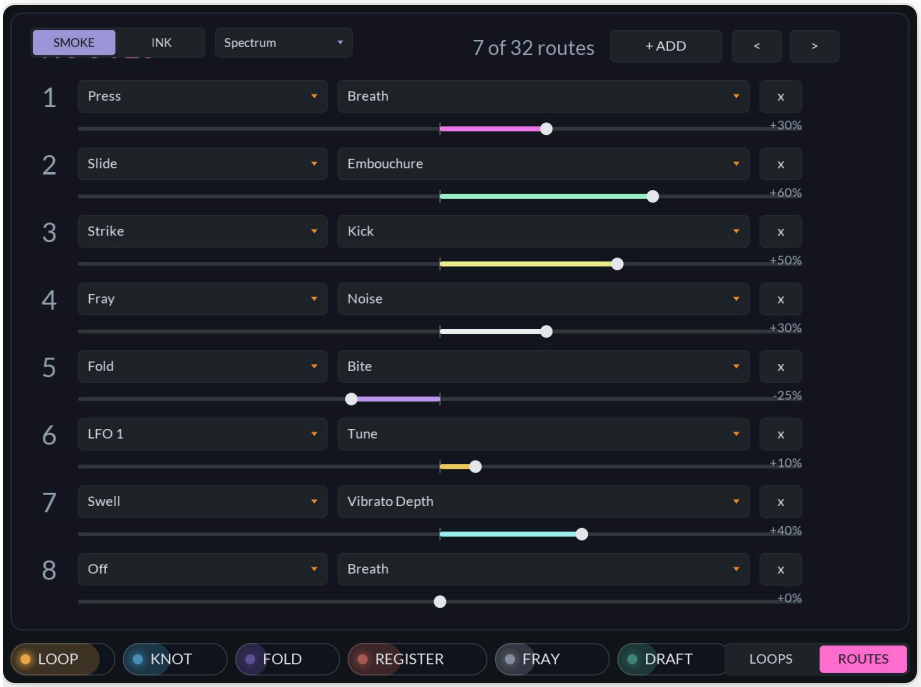


Fig. 11.2 ROUTES replaces the loops with the list of all 32 routes.

ROUTES, beside LOOPS under the stage, lists all 32 routes: each a source, a destination and an amount from −100% to +100%. Add, edit and remove them by hand here. A route's amount is a host parameter you can automate; its source and destination are not.

LFOs and Swell



Fig. 11.3 The LFOS tab on the player card. Each chip arms its source for MAP, like the chips in the header.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
LFO shape	`Sine, Triangle, Saw, Square, S&H, Smooth`	`Sine · Smooth`	LFO 1's and LFO 2's wave. Smooth is a gliding random.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
NOTE / FREE	`Note, Free`	`Note · Free`	Restart with each note, or run free, shared by every voice.
RATE	`0.02 – 20 Hz`	`1.00 · 0.25 Hz`	How fast. MOD ROUTE
Swell shape	`Arch, Rise, Fall, Waves, Bloom, Tide`	`Arch`	A rise and fall, a rise to stay, a fall, two humps, a quick bloom that settles, or a slow breath in and out.
ONCE / LOOP	`Once, Loop`	`Once`	Play the contour once per note, or keep repeating it.
TIME	`0.05 – 30 s`	`3.00 s`	How long the contour takes. MOD ROUTE

Buses



Fig. 11.4 The BUSES tab: four buses, each mixing what's routed to its AMOUNT and passing it on as a source.

A bus combines several sources into one. Route sources onto a bus's **AMOUNT**; the bus mixes them, bends the result and lags it, and sends it on as a source of its own (BUS 1 to 4). For example, PRESS and REGISTER into a bus set to Multiply, routed to GROWL, makes only loud, high notes growl.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Mode	`Sum, Multiply, Min, Max`	`Sum`	How the routes onto it combine.
Curve	`Linear, Exp, Log, S-Curve, Absolute`	`Linear`	Bends the mix.
Polarity	`As Is, Unipolar, Bipolar, Invert`	`As Is`	Shifts or flips it.
AMOUNT	`-100 – +100%`	`0%`	An offset, and where its inputs are routed. MOD ROUTE
LAG	`0 – 2000 ms`	`0 ms`	Smooths the output. MOD

CHAPTER 12

The Alien

An instrument that sings words, a syllable a note: yours, typed in, or words of its own made-up language.



Fig. 12.1 The ALIEN tab, singing typed words. Each word lights up while a voice sings it; the map shows where its vowels are (a human's hollow, its own lit).

Its voice is a formant synthesizer, a cascade of eight resonances for the vowels with a nose and a mouth beside them for the consonants, sung by vocal folds that shut faster and brighter the harder they're blown. Each syllable runs as a plan: the last syllable's closing consonants, this one's opening ones, then its vowel, held as long as the note is. Let go, it says the closing consonants before it falls silent. A chord's notes sing the same syllable; in LEGATO, a slur runs one syllable straight into the next. The breath is how loud and how hard it sings, the velocity how hard it says the consonants. It was tuned by playing whole sentences through the plug-in to speech recognizers.

Words

- **Typed words.** Type in the box and the source switches to **TYPED**. Words are looked up in the CMU Pronouncing Dictionary (125,000 words, built in); a word it hasn't got is read the way an English reader would read it, so made-up names work too. Phones in ARPAbet between brackets, such as `[HH AH0 L OW1]`, are sung as written. The words start from the top whenever the host starts playing, and are saved with the project.
- **Its own words.** With nothing typed it sings a made-up language: a vocabulary of a dozen words built from the vowels and consonant families that are lit, the commoner words coming back more often. **NEW WORDS**, or the **LANGUAGE** knob, picks another of a thousand languages. **SHORT / MIXED / LONG / CHANT** sets how long its words are.

The throat

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ACCENT	0 – 100%	0%	How far each syllable's vowel is bent from a human's, new each syllable. MOD ROUTE
SPREAD	-100 – +100%	0%	Squeezes the resonances together (a muffled hum) or pushes them apart (glassy and thin). MOD ROUTE
FLIP	0 – 100%	0%	Turns the vowels round the middle of the vowel space; all the way, ee is where ah was. MOD ROUTE
SIZE	-1 – +1 oct	+0.00 oct	Every resonance up or down an octave. A quarter up with the notes up an octave is a woman's voice; a quarter down a big man's. MOD ROUTE
SHARP	0 – 100%	50%	How narrow the resonances are, up to a throat singer's whistles. MOD ROUTE
WET	0 – 100%	20%	Something moving in the throat: the resonances shiver and gurgle. MOD ROUTE
ROUGH	0 – 100%	45%	How unsteady the folds are, as a living voice's. At 0% it's a machine. MOD ROUTE
GROWL	0 – 100%	0%	A creature's rasp: a growl an octave under the note, a rattle, a creak. MOD ROUTE
FOLLOW	0 – 100%	0%	How far the throat follows the note: none is a singer's. MOD ROUTE
WOBBLE	0 – 100%	0%	Each resonance swings on a slow wave of its own. MOD ROUTE
DRIFT	0 – 100%	20%	How far the vowels wander on their own. MOD ROUTE
GLIDE	5 – 400 ms	30 ms	How long each vowel takes to arrive. MOD ROUTE
BEND	0 – 100%	60%	How far the throat follows a pitch bend, like a talk box's mouth. MOD ROUTE
BREATHY	0 – 100%	25%	How much air goes through the folds, from clear to a whisper. MOD ROUTE
CONSONANT	0 – 100%	100%	How strong the consonants are, from vowels only to every stop, hiss and hum. MOD ROUTE
HOLD	0 – 100%	20%	How long a diphthong holds its first vowel. MOD ROUTE
TALK · PACE	0 – 100% · Free, 1/16 – 2 Bars	50% · Free	How soon a held note goes on to the next syllable, chattering while held. Free (by TALK; under a fifth, never), or a note value at the host's tempo. Typed words go on only with a PACE set (on Free, a syllable a note), and the next note picks up where the held one got to. MOD ROUTE
VOWELS · CONSONANTS	each On / Off	all On	Which its own words use: A E I O U, and M N, L R, W Y, V Z, B D G, H, F S, P T K.

The exciter on the Alien

EMBOUCHURE is the jaw, from nearly shut to wide open (slide opens the mouth). **STIFFNESS** is the vocal folds, from lax and hooting to pressed and buzzing. **BITE** sings through the nose below the middle and adds a brassy twang above it. **APERTURE** is the lips, from rounded and dark to a bright grin. **KICK** is how hard each word starts and its consonants hit, and **LOSS** on the bore a damped, muffled throat. Each changes its colour, not how loud it is.

The bowed string

BOW replaces the folds with a bowed string that sings the words: the breath is the bow's speed, BITE its weight, EMBOUCHURE where it crosses the string, STIFFNESS the hair's tension, APERTURE how wide it lies, and CONE how stiff the string is, from gut to a steel wire whose partials stretch apart like a bell's. Try *Talking Bow*.

High notes are sung as a singer sings them: from about E4 up, the first resonance rides above the note and the upper ones widen, so the top stays round rather than whistling. The ALIEN tab only shows when the Alien is the instrument.

CHAPTER 13

The Breath Loops and AIR View

Every voice drawn as a loop of its own sound, slowly turning, puffing smoke as it blows.

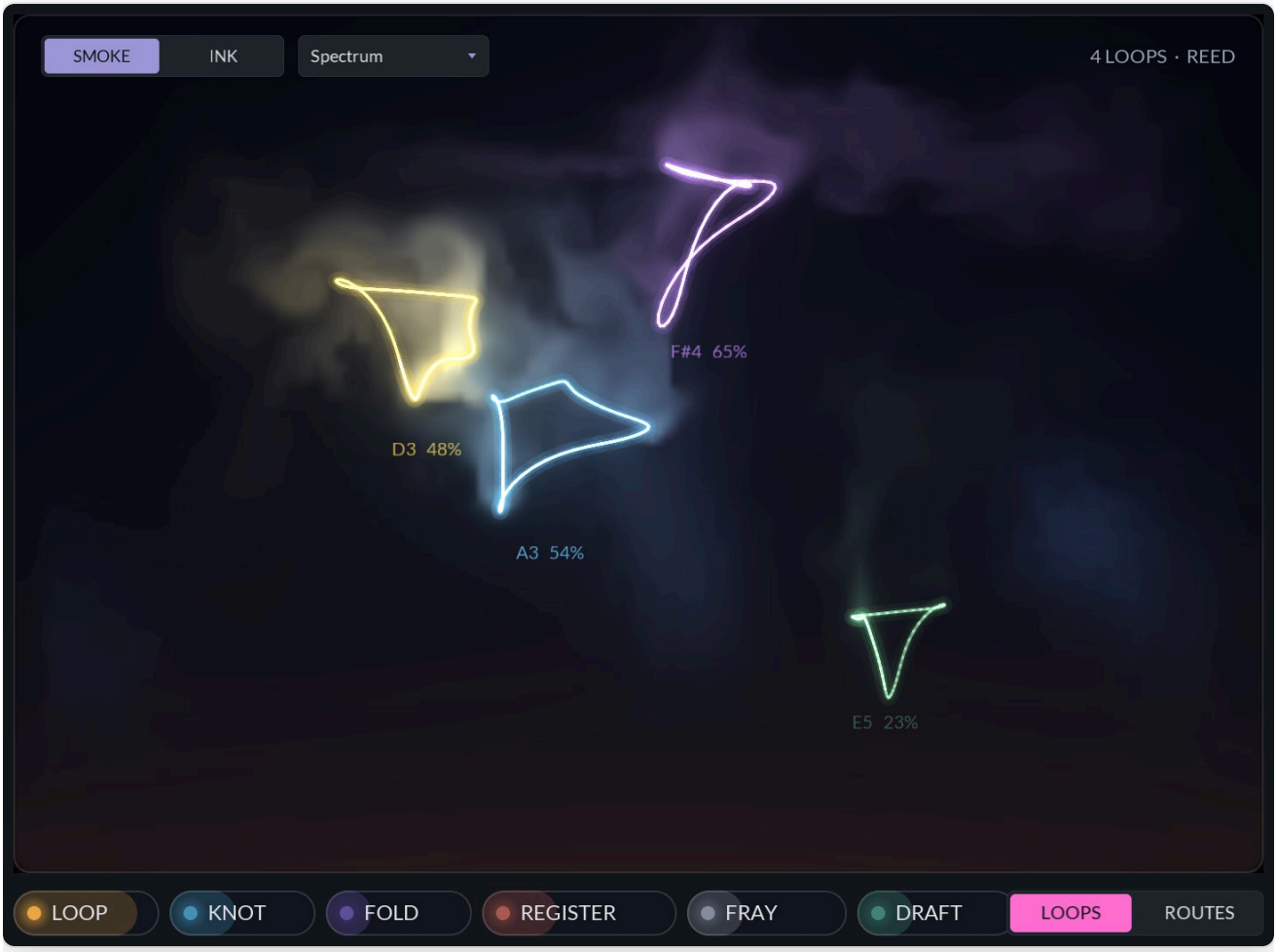


Fig. 13.1 *Smoke Tenor*, four notes. Each loop is labelled with its note and how hard it's blowing.

Reading a loop

Each loop plots a voice's sound now against its sound a quarter of a period ago and half a period ago, in three dimensions, slowly turning. A steady tone draws a clean closed loop, its shape the tone's harmonics; growl, flutter, breath noise or a split note fray it. That fraying is what the **FRAY** source measures. Smoke comes off every loop as it blows, in the loop's own colour, cooling to blue mist as it rises and curls away.

Above the stage, **SMOKE / INK** picks the look. **INK** drops dye from the loops that sinks and spreads instead of glowing smoke.

The menu beside it, **Colours**, picks the voices' colour scheme. Each loop glows, and its smoke or ink comes off it, in its voice's colour, and the same colour marks that note everywhere else on the panel. There are eight: **Spectrum** (the default, one of each hue), **Neon**, **Sunset**, **Ocean**, **Aurora**, **Pastel**, **Fire** and **Mono**. It replaces 0.12.0's smoke palette menu.

The AIR view

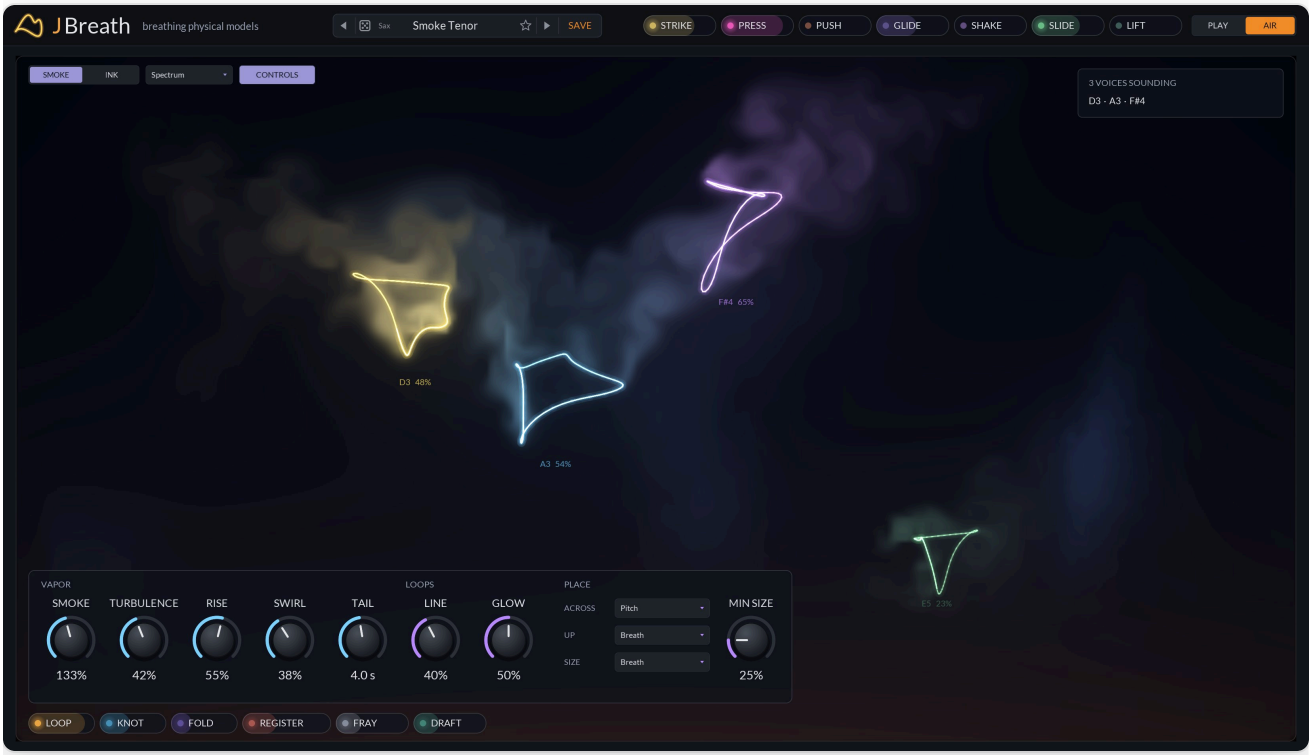


Fig. 13.2 The AIR view with its controls open.

AIR in the header gives the loops the whole window; **PLAY** brings the panel back. **CONTROLS** opens the look's knobs over the view. The loops show no tooltip here, so nothing covers them while you watch (the smaller loops card in PLAY keeps its tip).

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SMOKE	`0 – 300%`	`133%`	How much smoke the loops puff out.
TURBULENCE	`0 – 100%`	`42%`	How hard the air curls the smoke.
RISE	`0 – 100%`	`55%`	How fast it rises.
SWIRL	`0 – 100%`	`38%`	How tightly it curls into strands.
TAIL	`0.5 – 8.0 s`	`4.0 s`	How long it hangs.
LINE	`0 – 100%`	`40%`	The loops' line.
GLOW	`0 – 100%`	`50%`	How brightly loops and smoke are lit: 50% is the usual light, 100% four times it.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ACROSS · UP · SIZE	`Pitch, Strike, Press, Slide, Breath, Random, Centre`	`Pitch · Breath · Breath`	What places each loop across and up the stage, and what sets its size.
MIN SIZE	`10 – 100%`	`25%`	The smallest a loop gets, against the largest.

> **THE LOOK STAYS WITH THE PROJECT**
> The look is kept with the project, not with presets, so changing preset leaves it alone. Its knobs are not host parameters to automate.

The animation steps in time with the display and thins the smoke if frames run late, so the view stays smooth without taking time from the audio.

CHAPTER 14

The Voice Card

The voices, where they sit, and the room, the echo and the output they play into.



Fig. 14.1 The VOICE tab: the eight voices, where each note sits in the stereo field, and the meters.

Voice modes

MODE	VOICES	WHAT IT DOES
POLY	8	A ninth note takes the quietest released voice, or the oldest.
MONO	1	The tongue stops the air between notes. Letting go goes back to the newest key still held.
LEGATO	1	The fingers change and the air doesn't stop: a slur, with GLIDE setting how long the pitch takes to get there. The Double tongue style still tongues every note.

The VOICE tab

The tab shows the eight voices, lit while they sound; where each note sits in the stereo field, with the output's balance under it; and the meters, glowing as they rise: the breath coming in (**IN**), and the output left and right with the peak held (in dB at the top), red over 0 dB, with a notch where Master is set.

The knobs along the bottom

These stay on screen whichever tab is open.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SPREAD	-100 – +100%	50%	Low notes left, high right, as across a section; below zero the other way round. MOD
PAN	-100 – +100%	0%	Where the notes sit before the spread. Map Key, Slide or Random onto it to place each note. MOD ROUTE
UNISON	1 – 4	1	Copies of every voice, detuned against each other. MOD
U.DETUNE	0 – 50 ct	12 ct	How far the unison copies spread. MOD ROUTE
GLIDE	0 – 1000 ms	0 ms	How long the pitch takes to slide to a new note in LEGATO. MOD ROUTE
BEND	1 – 96 st	48 st	The MPE bend range of the note channels. MOD
TRANSP	-24 – +24 st	+0 st	Transpose. MOD ROUTE
REVERB	0 – 100%	15%	How much room. MOD
AIR	0 – 100%	0%	The breath heard around the instrument, not through it. MOD ROUTE
LEVEL	-60 – +12 dB	0.0 dB	The patch's level, saved with presets. MOD ROUTE
MASTER	-36 – +6 dB	0.0 dB	The output's level, before the limiter, so turning it up can't clip the output. The dice never touch it. MOD

The **AIR** tab (where the breath comes from) is described in Chapter 7.

The REVERB tab



Fig. 14.2 The REVERB tab, with the output's tone and the limiter.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
REVERB	Off, On	On	Switched off, the room's tail rings out rather than stopping.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SIZE	`0 – 100%`	`50%`	How big the room is. MOD
DECAY	`0.2 – 10.0 s`	`1.6 s`	How long it rings. MOD
DAMP	`0 – 100%`	`50%`	How fast the highs die. MOD
PRE-DELAY	`0 – 100 ms`	`12 ms`	Delay before the room answers. MOD
TONE	`-12 – +12 dB`	`+0.0 dB`	Darkens or brightens the whole output. MOD
LIMITER	`Off, Clip, Limit`	`Limit`	The very last stage, after Master. Off: nothing. Clip: overs saturate. Limit: a soft limiter that turns a loud chord down smoothly rather than letting it clip. It starts 5 dB under full scale and holds the loudest of it at −2 dBFS, with a soft knee, a 1 ms attack and a release that slows once it has been limiting a while, so a held chord doesn't pump; a soft clip after it catches the first instant of a loud note, so the output never reaches 0 dBFS. No latency, and anything under its knee passes untouched. (Projects saved with 0.12's Hard get this limiter.)

The DELAY tab



Fig. 14.3 The DELAY tab, synced to a dotted eighth.

A stereo echo before the reverb, off until switched on.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
DELAY	`Off, On`	`Off`	Switches the echo in.
SYNC	`Free, 1/32 – 1/1, dotted and triplet`	`Free`	A note value at the host's tempo, or Free to use TIME.
MIX	`0 – 100%`	`30%`	How loud the echoes are. MOD
TIME	`10 ms – 2 s`	`375 ms`	The echo time when SYNC is Free. MOD

CONTROL	RANGE	DEFAULT	WHAT IT DOES
FEEDBACK	0 – 95%	40%	How many echoes there are. MOD
TONE	0 – 100%	72%	How much darker each one gets. MOD
SPREAD	0 – 100%	50%	How far they bounce between left and right. MOD

CHAPTER 15

Presets and Dice

Eighty-seven factory presets, your own, and a die that builds new instruments.

The preset bar

- ◀ ▶ step to the previous or next preset, in menu order. Scrolling over the bar does the same.
- The **name** opens the preset menu: *Favorites*, the factory bank by family, a *User* submenu with your own presets, and *Save as…*. The family of the current preset is shown at the left of the name.
- ☆ beside the name makes the current preset a favorite; click it again to unstar it.
- **SAVE** asks for a name and writes the current settings as a user preset.
- The **die** at the left opens the dice card (below).

Loading a preset sets every parameter except the AIR view's look, so nothing else carries over from the previous sound. The host sees the change as an ordinary edit, so it can be undone.

User presets

User presets are small text files with the extension `.jbpreset`, one line per parameter. They live in:

macOS	`~/Library/Application Support/JBreath/Presets`
Windows	`%APPDATA%\JBreath\Presets`
Linux	`$XDG_CONFIG_HOME/JBreath/Presets` (or `~/.config/JBreath/Presets`)

Copy files into this folder to share presets between machines; they appear the next time the menu opens. Favorites are kept beside it and shared by every JBreath you open.

The factory bank

Init, then eight to twenty-seven presets for each family. Each family starts from its most natural sound, the model with a player's vibrato and a little room, and runs through darker, breathier, brighter and harder variants, its relatives by Transpose, and its oddities. Every factory preset plays in tune at a usable level (tested).

Flute · 10

From the concert flute to its relatives and a few oddities.

Concert Flute · Night Flute · Bamboo Air · Shakuhachi Bend · Tin Whistle · Piccolo · Alto Flute · Bass Flute · Overblown Jet (flutter-tongued) · Glass Ocarina

Clarinet · 9

A single reed on a cylinder, warm to raw.

Wooden Reed · Legato Clarinet · Chalumeau · Bass Clarinet · Bright Eb · Klezmer Cry · Breathy Reed · Low Rasp Reed · Felt Clarinet (a felt bucket over the bell)

Sax · 10

Alto, tenor, baritone and bass, clean to gutbucket.

Clean Alto · Smoke Tenor · Growl Alto · Subtone Ballad · Soprano Line · Baritone · Bass Honk · Gutbucket · Flutter Tenor · Hat Alto (a hat over the bell)

Oboe · 10

The double reeds and their relatives.

Dusk Oboe · Pastoral · Oboe d'Amore · Cor Anglais · Bassoon · Contra Buzz · Shawm · Musette · Duduk Air · Snake Charmer (a gourd cup over the reed)

Trumpet · 11

Open, the five mutes, and the brass family.

Brass Plume · Fanfare · Straight Mute · Cup Mute · Muted Night (Harmon) · Plunger Wah (plunger on the slide) · Bucket Ballad · Flugel Glow · French Horn · Trombone Slide · Piccolo Trumpet

Alien · 27

Voices that sing, speak and growl, for typed words or their own.

Alien Speech · First Contact (a plain voice for your words) · Talking Bow · Throat Singer · Xeno Choir · Tractor Beam (a word a beat) · Insect Hymn · Mothership · Signal Lost · Ion Flutter · Inside Out · Glass Mouth · Muffled Moan · Warble Choir · Tiny Giant · Swamp Creature · Clear Diction (most intelligible) · Crisp Diction · Chatterbox · Envoy Unit · Slow Drawl · Whisper Tongue · Hymn Singer · Ballad Voice · Folk Tenor · Baritone Chant · Lullaby

Pad · 6

The HYBRID voice as a pad under an instrument: low Follow, slow attack, long release.

Flute Halo · Hollow Pad · Section Pad · Reed Choir · Brass Swell · Nebula

Space · 6

The DELAY, synced and free, and the REVERB.

Echo Flute · Canyon Trumpet · Dub Tenor · Night Cathedral · Slapback Bari · Star Signal

Dice

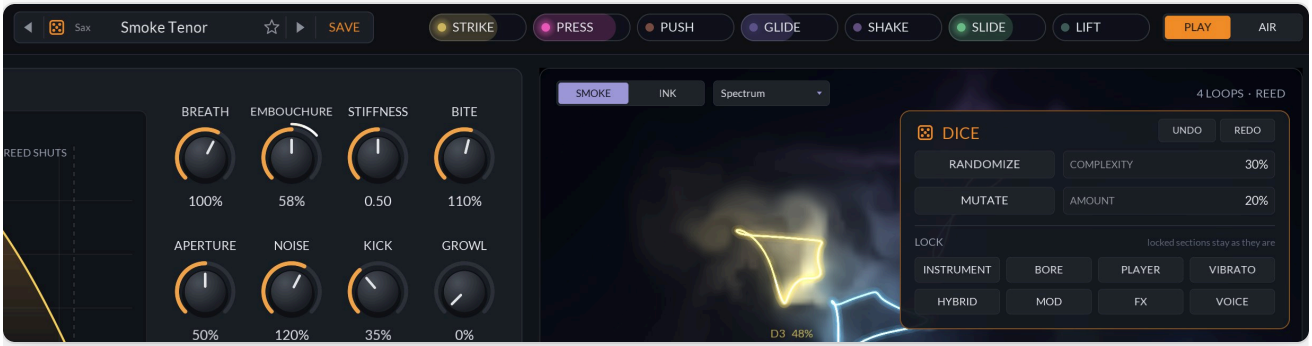


Fig. 15.1 The dice card opens over the breath loops, so nothing it changes is hidden.

RANDOMIZE builds a new patch the way a player would pick one up: an instrument first, then its exciter, bore, tongue and vibrato within the ranges that sound for that instrument, and a name for it. **COMPLEXITY** says how much it reaches for: at 0 a plain, real instrument; higher brings growl or flutter, a mute, the hybrid layer, unison, LFOs and routes, the delay, wider bore settings and the Alien.

MUTATE nudges the patch you have by **AMOUNT**: what's off stays off, a choice moves only to a near one (a mute to the next mute, a tongue to its neighbour), and only a high amount changes the instrument or adds a route. **UNDO** and **REDO** step through what the dice replaced.

LOCK keeps a section as it is through both: INSTRUMENT, BORE, PLAYER, VIBRATO, HYBRID, MOD, FX and VOICE. The dice never touch Level, Master or the Limiter, the breath source and the rest of the MIDI and MPE setup, Transpose, Tune, or the AIR view's look. The complexity, amount and locks stay with the plug-in while the project is open, not with presets. Every patch the dice make plays in tune at a usable level (tested over many seeds).

CHAPTER 16

Sound Design Recipes

Starting points for playing and for sound design. Begin from the preset named, then change only what is listed. A route is written SOURCE → destination, amount.

Breath-controller sax

A sax that brightens and buzzes as you lean in, from a breath controller or MPE pressure.

```
from Clean Alto
BREATH FROM Press · CURVE 0.80
PRESS → Bite, +40%
FOLD → Growl, +30%
```

Close to: Clean Alto, Growl Alto

Accents on the push

Pushing into a note bites; easing in stays round.

```
from Brass Plume
PUSH → Bite, +60%
PUSH → Kick, +30%
```

New sources in 0.12.0

Finger vibrato that growls

Shake the key and the tone roughens with the vibrato.

```
from Smoke Tenor
VIBRATO DEPTH 0%
SHAKE → Growl, +70%
```

Needs MPE pitch from the finger

Repeated notes on one key

Tongue a held trumpet note again with a quick dip in pressure, as a player does on one fingering.

```
from Fanfare
RE-TONGUE 80% · STYLE Hard
FORCE Press
```

Close to: Fanfare

Plunger wah

Slide works the plunger like a hand over the bell.

```
from Brass Plume
MUTE Plunger · MUTE AMT 0%
SLIDE → Mute Amount, +100%
```

Close to: Plunger Wah

Section pad

A small reed section with a soft sine under it, swelling in after each note.

```
from Wooden Reed
UNISON 3 · U.DETUNE 14 ct · DETUNE 8 ct
HYBRID Sine -1 oct · MIX 30% · FOLLOW 20%
HYBRID ATTACK 800 ms · RELEASE 2 s
```

Close to: Reed Choir, Section Pad

Only loud high notes growl

Two sources multiplied on a bus.

```
from Growl Alto · GROWL 0%
PRESS → Bus 1, +100% · REGISTER → Bus 1, +100%
BUS 1 Multiply
BUS 1 → Growl, +80%
```

Uses a bus as an AND

Overblow latch

Once a note is blown past the top it stays there until you let go. Feedback on purpose.

```
from Concert Flute
BREATH 120%
FOLD → Breath, +50%
```

Back off the amount if it runs away

General tips

- **Blow the dynamics, don't mix them.** A wind instrument's loudness and tone are the same control. Use the breath to play soft and loud, and LEVEL only to balance the patch.
- **Lower Breath for a beginner, raise it for a pro.** BREATH under 100% never quite reaches the instrument's top; over 100% every note can be overblown.
- **Play in range.** Each model is best in its own register. Use TRANSP or a relative preset rather than playing a flute two octaves down.
- **Legato for lines.** LEGATO with a little GLIDE sounds like one player; POLY sounds like a section.
- **Humanize a little.** 10 to 20% HUMANIZE and a little DETUNE make repeated notes sound played, not pasted.

APPENDIX A

Parameter Reference

Every parameter JBreath exposes to the host, grouped as the host lists them. Everything but the AIR view's look is automatable.

EXCITER

PARAMETER	RANGE	DEFAULT	
Instrument	Flute, Clarinet, Sax, Oboe, Trumpet, Alien	Clarinet	
Breath	25 – 150%	100%	MOD ROUTE
Embouchure	0 – 100%	50%	MOD ROUTE
Stiffness	0.00 – 1.00	0.50	MOD ROUTE
Bite	0 – 200%	100%	MOD ROUTE
Aperture	0 – 100%	50%	MOD ROUTE
Noise	0 – 200%	100%	MOD ROUTE
Kick	0 – 100%	25%	MOD ROUTE
Growl	0 – 100%	0%	MOD ROUTE

ARTICULATION

PARAMETER	RANGE	DEFAULT	
Tongue	Soft, Hard, Double, Flutter, Breath	Hard	
Attack	1 – 400 ms	25 ms	MOD ROUTE
Release	5 – 2000 ms	90 ms	MOD ROUTE
Humanize	0 – 100%	0%	MOD ROUTE
Flutter	0 – 100%	0%	MOD ROUTE
Flutter Rate	10 – 40 Hz	24.0 Hz	MOD ROUTE
Re-Tongue	0 – 100%	50%	MOD ROUTE

VIBRATO

PARAMETER	RANGE	DEFAULT	
Vibrato Type	Breath, Jaw, Breath+Jaw, Lip	Breath	
Vibrato Rate	2 – 9 Hz	5.10 Hz	MOD ROUTE
Vibrato Depth	0 – 30%	5%	MOD ROUTE
Vibrato Pitch	0 – 50 ct	0 ct	MOD ROUTE
Vibrato Delay	0 – 2000 ms	350 ms	MOD ROUTE

BORE

PARAMETER	RANGE	DEFAULT	
Tune	-100 – +100 ct	+0 ct	MOD ROUTE
Glide	0 – 1000 ms	0 ms	MOD ROUTE
Length	0.15 – 1.50 m	0.67 m	MOD
Loss	0 – 50%	4%	MOD ROUTE
Hole Cutoff	0.50 – 4.00 kHz	1.30 kHz	MOD ROUTE
Cone	0 – 100%	50%	MOD ROUTE
Radiate	0 – 100%	66%	MOD
Brass	0 – 100%	50%	MOD ROUTE
Barrel	0 – 100%	40%	MOD
Wall	Metal – Bone	Wood	MOD
Detune	0 – 25 ct	+0 ct	MOD ROUTE
Holes	Closed, Lattice 3/5/7/9, Open	Lattice 7	
Tube	Grenadilla … Glass (8)	Grenadilla	
Mute	Open, Straight, Cup, Harmon, Plunger, Bucket	Open	
Mute Amount	0 – 100%	100%	MOD ROUTE

VOICE

PARAMETER	RANGE	DEFAULT	
Voice Mode	Poly, Mono, Legato	Poly	
Breath Source	Auto, Pressure, Velocity, Input, Mix	Auto	
Tongue Force	Strike, Press	Strike	
Press Curve	0.30 – 3.00	1.00	MOD ROUTE
Slide	0 – 100%	50%	MOD
MPE Bend Range	1 – 96 st	48 st	MOD
Breath In	0 – 100%	0%	MOD
Transpose	-24 – +24 st	+0 st	MOD ROUTE
Unison	1 – 4	1	MOD
Unison Detune	0 – 50 ct	12 ct	MOD ROUTE

HYBRID

PARAMETER	RANGE	DEFAULT	
Hybrid Wave	Sine, Triangle, Saw, Square, Pulse	Saw	
Hybrid Mix	0 – 100%	0%	MOD ROUTE
Hybrid Octave	-2 – +2 oct	+0 oct	MOD

PARAMETER	RANGE	DEFAULT	
Hybrid Detune	-50 – +50 ct	+0 ct	MOD ROUTE
Hybrid Breath	50 – 150%	100%	MOD
Hybrid Cutoff	80 Hz – 12 kHz	1.80 kHz	MOD ROUTE
Hybrid Resonance	0 – 100%	20%	MOD ROUTE
Hybrid Filter Breath	0 – 100%	60%	MOD ROUTE
Hybrid Attack	1 ms – 10 s	5.2 ms	MOD
Hybrid Decay	1 ms – 10 s	251 ms	MOD
Hybrid Sustain	0 – 100%	100%	MOD
Hybrid Release	1 ms – 10 s	63 ms	MOD
Hybrid Follow	0 – 100%	100%	MOD
Hybrid Filter Env	-100 – +100%	0%	MOD

OUTPUT, ROOM AND DELAY

PARAMETER	RANGE	DEFAULT	
Spread	-100 – +100%	50%	MOD
Pan	-100 – +100%	0%	MOD ROUTE
Air	0 – 100%	0%	MOD ROUTE
Level	-60 – +12 dB	0.0 dB	MOD ROUTE
Master	-36 – +6 dB	0.0 dB	MOD
Limiter	Off, Clip, Limit	Limit	
Reverb	Off, On	On	
Room	0 – 100%	15%	MOD
Room Size	0 – 100%	50%	MOD
Room Decay	0.2 – 10.0 s	1.6 s	MOD
Room Damping	0 – 100%	50%	MOD
Room Pre-Delay	0 – 100 ms	12 ms	MOD
Tone	-12 – +12 dB	+0.0 dB	MOD
Delay	Off, On	Off	
Delay Sync	Free, 1/32 – 1/1	Free	
Delay Mix	0 – 100%	30%	MOD
Delay Time	10 ms – 2 s	375 ms	MOD
Delay Feedback	0 – 95%	40%	MOD
Delay Tone	0 – 100%	72%	MOD

PARAMETER	RANGE	DEFAULT	
Delay Spread	0 – 100%	50%	MOD

MODULATION

PARAMETER	RANGE	DEFAULT	
LFO 1 Rate · Shape · Sync	0.02 – 20 Hz · 6 shapes · Note, Free	1.00 Hz · Sine · Note	MOD ROUTE
LFO 2 Rate · Shape · Sync	0.02 – 20 Hz · 6 shapes · Note, Free	0.25 Hz · Smooth · Free	MOD ROUTE
Swell Shape · Time · Loop	6 shapes · 0.05 – 30 s · Once, Loop	Arch · 3.00 s · Once	MOD ROUTE
Bus 1–4 Amount	-100 – +100%	0%	MOD ROUTE
Bus 1–4 Mode · Curve · Polarity	Sum, Multiply, Min, Max · 5 curves · 4 polarities	Sum · Linear · As Is	
Bus 1–4 Lag	0 – 2000 ms	0 ms	MOD
Route 1–32 Source · Destination	22 sources · 62 destinations	Off	
Route 1–32 Amount	-100 – +100%	0%	MOD

ALIEN

PARAMETER	RANGE	DEFAULT	
Alien Accent	0 – 100%	0%	MOD ROUTE
Alien Wobble	0 – 100%	0%	MOD ROUTE
Alien Drift	0 – 100%	20%	MOD ROUTE
Alien Shift (Size)	-1 – +1 oct	+0.00 oct	MOD ROUTE
Alien Sharp	0 – 100%	50%	MOD ROUTE
Alien Bow	0 – 100%	0%	MOD ROUTE
Alien Talk	0 – 100%	50%	MOD ROUTE
Alien Pace	Free, 1/16 – 2 Bars	Free	
Alien Hold	0 – 100%	20%	MOD ROUTE
Alien Glide	5 – 400 ms	30 ms	MOD ROUTE
Alien Bend	0 – 100%	60%	MOD ROUTE
Alien Consonants	0 – 100%	100%	MOD ROUTE
Alien M N … P T K (8)	Off, On	On	
Alien A … U (5)	Off, On	On	
Alien Word Length	Short, Mixed, Long, Chant	Mixed	
Alien Breathy	0 – 100%	25%	MOD ROUTE
Alien Spread	-100 – +100%	0%	MOD ROUTE

PARAMETER	RANGE	DEFAULT	
Alien Flip	0 – 100%	0%	MOD ROUTE
Alien Follow	0 – 100%	0%	MOD ROUTE
Alien Words	Alien, Typed	Alien	
Alien Language	0 – 999	1	MOD
Alien Rough	0 – 100%	45%	MOD ROUTE
Alien Growl	0 – 100%	0%	MOD ROUTE
Alien Wet	0 – 100%	20%	MOD ROUTE

VIEW (NOT AUTOMATABLE, NOT IN PRESETS)

PARAMETER	RANGE	DEFAULT
Look	Smoke, Ink	Smoke
Colours	Spectrum, Neon, Sunset, Ocean, Aurora, Pastel, Fire, Mono	Spectrum
Smoke	0 – 300%	133%
Turbulence	0 – 100%	42%
Rise	0 – 100%	55%
Swirl	0 – 100%	38%
Tail	0.5 – 8.0 s	4.0 s
Line	0 – 100%	40%
Glow	0 – 100%	50%
Loop X · Y · Size	Pitch, Strike, Press, Slide, Breath, Random, Centre	Pitch · Breath · Breath
Loop Size Min	10 – 100%	25%

APPENDIX B

Specifications

Format	CLAP 1.2 instrument. Features: instrument, synthesizer, stereo. CLAP only: no VST3 or AU. Not supported in Ableton Live.
Notes	CLAP notes, MIDI and MIDI MPE. CLAP note expressions: tuning, pressure, brightness. Polyphonic aftertouch, channel pressure, CC 2 (breath), CC 74 (slide), CC 64 (sustain), RPN 0 (bend range), MPE Configuration Message.
Voices	8, with up to 4 unison copies each. Poly, Mono and Legato.
Output	Stereo.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit).
Synthesis	Digital waveguides with nonlinear exciters: air jet, single reed, cane reed, double reed, lips, vocal folds or bowed string over a formant synthesizer.
Internal rate	88.2 kHz or more (2× at 44.1 and 48 kHz), decimated once.
Tuning	Every note of every model within a cent, measured offline and corrected for breath.
Modulation	22 sources, 32 routes, 4 buses, 2 LFOs and a Swell contour, per voice.
Effects	Stereo delay (10 ms to 2 s, synced or free), room reverb, output tone, limiter.
Tail	Reported to the host from the release, the room and the delay.
Presets	87 factory presets; user presets as `.jbpreset` text files.
State	Saved with the host project, including the preset name, the Alien's typed words and the AIR view's look.
Panel	1648 × 940 at 100%, scalable.

Troubleshooting

SYMPTOM	WHAT TO CHECK
JBreath doesn't appear in my host	Your host must load CLAP plug-ins; Ableton Live does not. On macOS the download may be quarantined: see the warning in Chapter 2, then rescan.
Notes don't sound, or only on hard hits	With BREATH FROM at PRESS or INPUT, a note with no pressure or Breath In blows nothing. Set AUTO. Check BREATH isn't turned right down, and that CURVE isn't very high for a light-touch controller.
Pressure does nothing	Check your host passes aftertouch or MPE to the plug-in, and that BREATH FROM is AUTO, PRESS or MIX.
Bends are far too wide or too narrow	BEND on the voice card must match your controller's MPE bend range (48 is the standard). A controller that sends RPN 0 sets it for you.
A note splits, squeals or jumps a register	That's overblowing, as on the real instrument. Lower BREATH, or EMBOUCHURE on the flute; check for a breath source routed to Breath.

SYMPTOM	WHAT TO CHECK
Repeated notes when I didn't mean them	RE-TONGUE is tonguing quick dips in your pressure. Turn it down, or to 0%.
A patch runs away or latches loud	A breath source (LOOP, FOLD, KNOT, FRAY) is routed to Breath and feeding back. Open ROUTES and lower it.
It sounds out of tune	Check TUNE, TRANSP, DETUNE, U.DETUNE, HUMANIZE and the vibrato's PITCH. Every note is in tune with these at zero.
My user presets are missing	Check the preset folder for your OS (Chapter 15). Files must end in `.jbpreset`.

Glossary

Bore	The tube the sound rings in, modelled as delay lines carrying the wave down and back.
Exciter	The part the player blows into: a reed, the lips, an air jet or vocal folds. A valve the breath drives.
Waveguide	A way of modelling a tube or string with delay lines and filters, so the sound travels in it as in the real one.
Impedance	How hard the bore resists air at each frequency. Its peaks are the notes it can play.
Overblow	Blowing hard enough that the instrument jumps to a higher resonance: a split note or the next register.
Embouchure	The lips, jaw and the way they meet the instrument.
MPE	MIDI Polyphonic Expression: each note on its own channel, so each has its own bend, pressure and slide.
Formant	A resonance of the throat and mouth. Vowels are told apart by where their formants are.
Route	A link from a modulation source to a knob, with an amount.

JBreath

BREATHING PHYSICAL MODELS · V0.13.3 DRAFT

github.com/jeffmhopkins/Jclap